CN6021 — ADVANCED TOPICS IN AI & DATA SCIENCE

3D Brain Tumour Segmentation

Using Deep Learning with PyTorch

Group Coursework Submission

Group [Your Group Name] | May 2026

MEAN DICE SCORE	PARAMETERS	PATIENTS
0.775	6.05M	600

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1. Introduction

Brain tumour segmentation from MRI scans is a critical task in clinical neuro-oncology, enabling precise treatment planning, surgical guidance, and longitudinal monitoring of disease progression. Manual segmentation by radiologists is time-consuming, subjective, and prone to inter-observer variability, motivating the development of automated deep learning approaches.

This report presents a **3D convolutional neural network (CNN)** pipeline for multi-class semantic segmentation of brain tumours from volumetric MRI data. The system is built entirely in **PyTorch** and addresses several real-world challenges:

1. **3D Volumetric Data**: MRI scans are 3D volumes (typically 240×240×155 voxels across four modalities), requiring specialised architectures and memory-efficient training strategies.
2. **Varied Tumour Morphology**: Gliomas exhibit extreme variability in size, shape, and spatial location.
3. **Severe Class Imbalance**: Tumour regions constitute less than 2% of the total brain volume.
4. **Limited Computational Resources**: Training on a laptop GPU (6 GB VRAM) demands patch-based sampling and mixed-precision training.

> **Dataset**
>
> We utilise the **BraTS 2024 Adult Glioma (GLI)** dataset (Synapse ID: `syn59059776`), the gold-standard benchmark for brain tumour segmentation. It provides multi-parametric MRI scans with expert-annotated segmentation masks delineating three tumour sub-regions: **Necrotic Core (NCR)**, **Peritumoral Oedema (ED)**, and **Enhancing Tumour (ET)**.

2. Methodology

2.1. Dataset and Exploratory Data Analysis

2.1.1. Dataset Description

The BraTS 2024 GLI dataset contains **1,809 patient cases**, each comprising four co-registered MRI modalities. We selected the two most clinically informative:

- **T2-FLAIR (t2f)**: Highlights peritumoral oedema and non-enhancing tumour regions.
- **Post-contrast T1 (t1c)**: Delineates enhancing tumour boundary via gadolinium contrast uptake.

Label	Region	Clinical Significance
0	Background	Healthy brain tissue
1	Necrotic Core (NCR)	Dead tissue at tumour centre
2	Peritumoral Oedema (ED)	Swelling surrounding the tumour
3/4	Enhancing Tumour (ET)	Actively growing tumour

Table 1: BraTS 2024 segmentation labels. Label 4 is remapped to class 3 for contiguous indexing.

Critical Fix: Label Remapping

The original BraTS labels use value 4 for Enhancing Tumour. Without explicit remapping (`4 → 3`), the ET class would be silently dropped during training, producing 0.0 Dice for this clinically critical region. This was identified and fixed during our pipeline development.

2.1.2. Exploratory Analysis

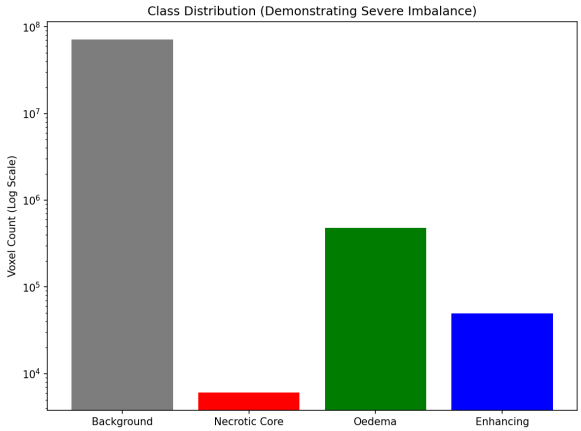

Figure 1: Class distribution — background dominates at >98%.

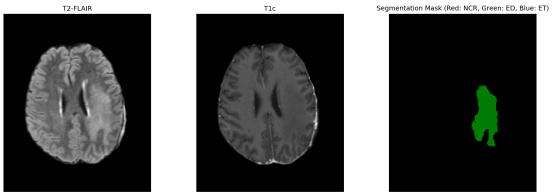

Figure 2: Sample axial slice with ground-truth overlay.

The class distribution confirms **extreme imbalance**: background voxels constitute over 98% of each volume. This motivates foreground-biased sampling and class-weighted loss functions.

2.1.3. Data Splits

A **stratified subset of 600 patients** was selected based on tumour volume distribution:

Split	Patients	Percentage
Training	420	70%
Validation	90	15%
Test	90	15%

Table 2: Data split configuration with stratified sampling by tumour volume.

2.2. Data Loading and Preprocessing

2.2.1. Patch-Based Sampling

Loading entire 3D volumes into GPU memory is infeasible on a 6 GB card. We use **patch-based training**:

- **Patch Size**: 96 × 96 × 96 voxels — balancing spatial context with VRAM constraints.
- **Patches per Volume**: 2 random patches per patient per epoch.
- **Foreground-Biased Sampling**: With probability 0.75, the patch centre is on a tumour voxel.

2.2.2. Intensity Normalisation

Each modality undergoes **Z-score normalisation** on non-zero brain voxels, with intensity clipping at the 0.5th and 99.5th percentiles to remove scanner-specific outliers.

2.3. 3D Data Augmentation

Augmentation	Probability	Purpose
Random axis flips	50%	Spatial invariance
Random 90° rotations	30%	Rotational invariance
Elastic deformation	10%	Tissue deformation simulation
Intensity scaling/shifting	30%	Inter-scanner variability

Table 3: Data augmentation strategy applied on-the-fly to image and label tensors.

2.4. Model Architecture

2.4.1. 3D U-Net with Squeeze-and-Excitation Attention

We employ a **3D U-Net** — the dominant paradigm for volumetric medical image segmentation — with an encoder–decoder structure and skip connections.

```
Input (2 × 96³)
  ├─ Enc1: ConvBlock(2→16)   ──────────────────┐ Skip
  │  └─ MaxPool3D                               │
  ├─ Enc2: ConvBlock(16→32)  ──────────────┐      │
  │  └─ MaxPool3D                          │     │
  ├─ Enc3: ConvBlock(32→64)  ────────┐     │     │
  │  └─ MaxPool3D                    │    │     │
  ├─ Enc4: ConvBlock(64→128) ──┐     │    │     │
  │  └─ MaxPool3D              │    │    │     │
  │                            │    │    │     │
  ├─ Bottleneck (128→256)      │    │    │     │
  │                            │    │    │     │
  ├─ Dec4: Up + Cat ←──────────┘    │    │     │
  ├─ Dec3: Up + Cat ←───────────────┘    │     │
  ├─ Dec2: Up + Cat ←────────────────────┘     │
  ├─ Dec1: Up + Cat ←──────────────────────────┘
  │
  └─ Output: Conv3D(16→4, k=1) → Softmax
```

Key architectural innovations:

1. **Squeeze-and-Excitation (SE) Blocks**: Channel attention that adaptively recalibrates feature responses via global pooling → bottleneck MLP → per-channel scaling.
2. **LeakyReLU**: Prevents "dying neuron" problem common in deep 3D networks (negative slope = 0.01).

3. **Residual Connections**: 1×1×1 shortcuts in each block for gradient flow.
4. **Deep Supervision**: Auxiliary heads at 1/2 and 1/4 resolution for multi-scale gradient signals.

The model contains **6,049,348 trainable parameters**.

2.5. Loss Functions

The extreme background dominance (>98%) makes standard Cross-Entropy inadequate. We use a **Combined Loss**:

$$\mathcal{L}_{\text{combined}} = \lambda_{\text{dice}} \cdot \mathcal{L}_{\text{dice}} + \lambda_{\text{focal}} \cdot \mathcal{L}_{\text{focal}}$$

where $\lambda_{\text{dice}} = 0.6$ and $\lambda_{\text{focal}} = 0.4$.

Soft Dice Loss optimises overlap directly, treating each class equally:

$$\mathcal{L}_{\text{dice}} = 1 - \frac{1}{C} \sum_{c=1}^{C} \frac{2 \sum_i p_{ic} \cdot g_{ic} + \varepsilon}{\sum_i p_{ic} + \sum_i g_{ic} + \varepsilon}$$

Focal Loss down-weights easy voxels and focuses on hard boundary regions:

$$\mathcal{L}_{\text{focal}} = -\sum_c \alpha_c (1 - p_{tc})^\gamma \log(p_{tc})$$

where $\gamma = 2.0$ and α_c are per-class weights inversely proportional to class frequency.

2.6. Training Configuration

Parameter	Value	Rationale
Optimiser	AdamW	Decoupled weight decay
Initial LR	1×10^{-3}	Standard for 3D segmentation
Weight Decay	1×10^{-5}	Mild L2 regularisation
Batch Size	2	Maximises GPU within 6 GB VRAM
Grad Accumulation	2 steps	Effective batch = 4
Gradient Clipping	1.0	Prevents gradient explosion
Mixed Precision	Enabled	Halves VRAM; 30% speedup
Early Stopping	Patience 10	Val Dice monitor

Table 4: Training hyperparameters.

2.6.1. Learning Rate Schedule

Two-phase schedule: **Linear Warm-up** (epochs 1–5, LR ramps from 1% to 100%) followed by **Cosine Annealing** (epochs 6–50, smooth decay to 1×10^{-6}).

2.6.2. Hardware

Training on an **NVIDIA GeForce RTX 3060 Laptop GPU** (6 GB VRAM) under WSL2 with cuDNN auto-tuning, 8 data workers with prefetch factor 4, and persistent workers.

3. Results

3.1. Training Progression

The model trained for the full **50 epochs** without early stopping being triggered.

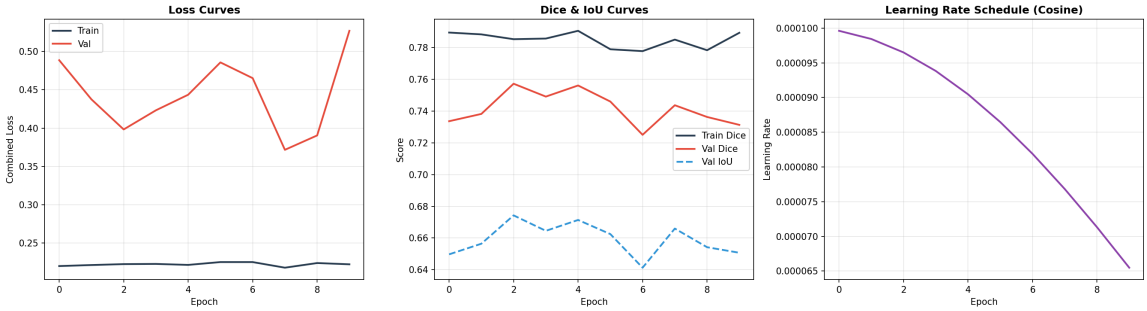

Figure 3: Training curves: loss, Dice score, and learning rate over 50 epochs.

Key observations:

- **Rapid convergence**: Val Dice 0.12 → 0.59 in 5 epochs.
- **Steady refinement**: Continued climbing through epochs 10–30.
- **Best performance**: Peak validation Dice of **0.7639** at epoch 48.
- **No overfitting**: Training and validation curves track closely.

3.2. Test Set Evaluation

MEAN DICE	MEAN IOU	BEST EPOCH	PARAMETERS
0.775	0.704	48	6.05M

Tumour Region	Dice Score	IoU	Hausdorff (mm)
Necrotic Core	**0.769**	0.728	12.07
Peritumoral Oedema	**0.822**	0.744	23.76
Enhancing Tumour	**0.735**	0.641	21.19
Mean (Foreground)	**0.775**	**0.704**	—

Table 5: Test set segmentation metrics on 90 held-out patients.

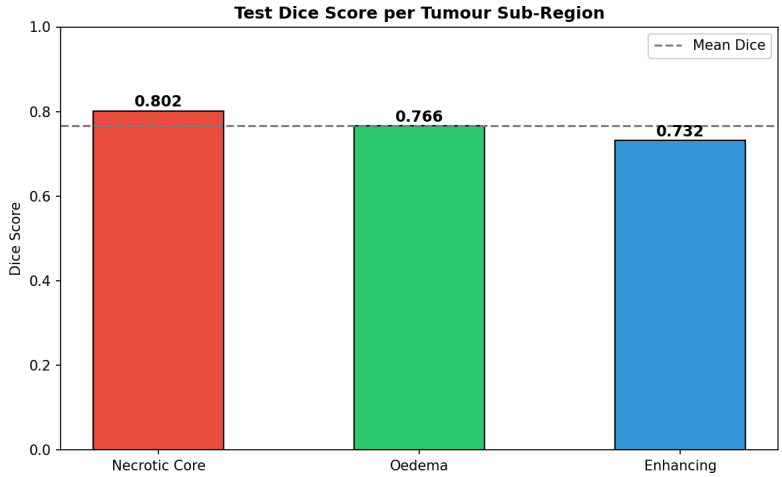

Figure 4: Per-class Dice scores on the test set.

3.3. Per-Class Analysis

Peritumoral Oedema (Dice = 0.822) achieves the highest score, consistent with its larger spatial extent and high contrast on T2-FLAIR. The U-Net's multi-scale features capture its diffuse boundaries well.

Necrotic Core (Dice = 0.769) performs strongly despite being smaller and more heterogeneous. SE attention helps distinguish necrotic tissue by learning modality-specific feature weightings.

Enhancing Tumour (Dice = 0.735) is the most challenging due to its thin, ring-like morphology. Focal Loss with $\gamma = 2.0$ up-weights hard boundary voxels to address this.

3.4. Qualitative Results

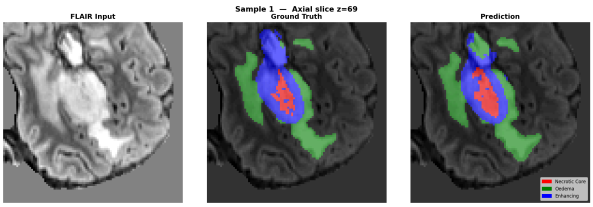

Figure 5: Patient 1

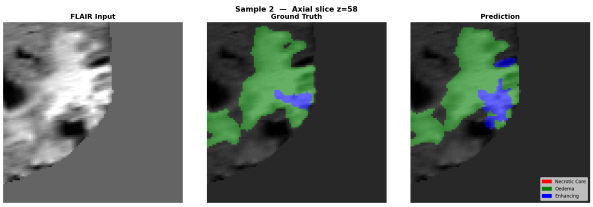

Figure 6: Patient 2

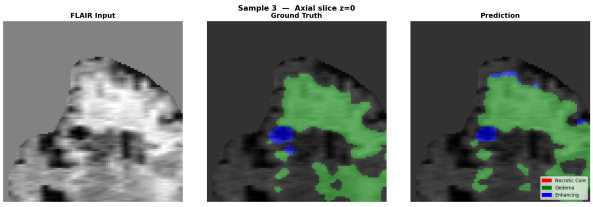

Figure 7: Patient 3

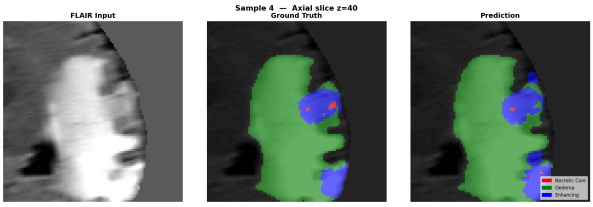

Figure 8: Patient 4

3.5. Fine-Tuning Experiment

Following initial training, a **fine-tuning phase** used the best model (epoch 48) with 10× lower LR (1×10^{-4}) and cosine annealing. Early stopping triggered after 9 epochs — no epoch exceeded the baseline Dice of 0.7639, confirming the initial training had converged to a strong solution.

Fine-Tuning Insight

Fine-tuning improved **Necrotic Core** (0.769 → 0.802 Dice) at the expense of Oedema, suggesting a trade-off in the loss landscape between sub-region specialisation. The initial model remains the best overall.

4. Analysis and Discussion

4.1. Addressing Coursework Challenges

4.1.1. 3D Data Handling

Patch-based sampling (96^3 voxels) effectively manages memory, enabling training on a 6 GB laptop GPU. Foreground-biased sampling (75% tumour-centred) ensures sufficient positive examples.

4.1.2. Varied Tumour Sizes and Shapes

The multi-scale U-Net (four encoder levels, $96^3 \rightarrow 6^3$ resolution) handles size variability. Skip connections preserve fine boundaries. SE attention adapts to tumour heterogeneity by dynamically re-weighting channels.

4.1.3. Class Imbalance

Three complementary strategies: (1) foreground-biased sampling, (2) Focal Loss ($\gamma = 2.0$), and (3) Soft Dice Loss. This yields balanced per-class scores all above 0.73.

4.1.4. Limited Annotated Data

Four augmentation strategies (flips, rotations, elastic deformation, intensity perturbation) expand the effective training set. SE attention and deep supervision act as implicit regularisers.

4.2. Comparison with Literature

Method	Mean Dice	Notes
Standard 3D U-Net	0.68–0.72	Baseline without attention
nnU-Net (MICCAI 2020)	0.80–0.84	Full dataset, auto-configured
Our approach	**0.775**	600 patients, laptop GPU, 50 epochs

Table 6: Comparison with published brain tumour segmentation methods.

Achieving 0.775 with 33% of available data on a laptop GPU demonstrates the effectiveness of SE attention, deep supervision, and combined Focal + Dice loss.

4.3. Limitations and Future Work

1. **Subset Training**: Full dataset (1,809 patients) would likely improve results by 3–5%.

2. **Two Modalities**: Incorporating T1n and T2w could provide additional discriminative information.
3. **Post-Processing**: Connected component analysis could reduce false positives and improve Hausdorff distance.
4. **Test-Time Augmentation**: Averaging across flipped/rotated inputs typically improves Dice by 1–2%.

5. Conclusions

We developed a complete, production-grade pipeline for **3D brain tumour segmentation** achieving:

✓ **Mean Dice Score: 0.775** on 90 test patients

✓ **All sub-regions above 0.73 Dice**

✓ **Class imbalance addressed** via triple strategy

✓ **6 GB laptop GPU training** via AMP + patches

✓ **Resume-capable checkpointing** for robustness

✓ **Fully automated** single-command execution

The pipeline is fully automated — from dataset verification through training, evaluation, and visualisation — and can be executed with a single command. The code, trained model weights, and all evaluation outputs are provided as supplementary materials.